# Exponential functions

**1** Consider the function $y = 5 \times 4^x$. Find the value of $y$ when:

a $x = 0$

b $x = 2$

c $x = -1$

d $x = 3$

**2** Consider the function $f(x) = \left(\dfrac{4}{3}\right)^x$.

a Evaluate $f(0)$.

b Evaluate $f(-2)$.

**3** Find the value of $5000(1.08^x)$ for $x = 30$ to two decimal places.

**4** Find the missing coordinate in each ordered pair that is a solution to $y = 5^x$:

a $(3, \square)$

b $(\square, 5)$

c $(-1, \square)$

d $\left(\square, \dfrac{1}{25}\right)$

**5** Find the missing coordinate in each ordered pair that is a solution to $y = -2^x$:

a $(5, \square)$

b $\left(\square, -\dfrac{1}{8}\right)$

c $(-1, \square)$

d $(\square, -16)$

**6** If the point $(3, m)$ lies on the curve $y = 2^x$, solve for $m$.

**7** The points $(3, n)$, $(k, 16)$ and $\left(m, \dfrac{1}{4}\right)$ all lie on the curve with equation $y = 2^x$.

a Solve for $n$.

b Solve for $k$.

c Solve for $m$.

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# Characteristics of exponential functions

**8** Do either of the functions $y = 9^x$ or $y = 9^{-x}$ have $x$-intercepts? Explain your answer.

**9** For what values of $x$ will $6^x$ have a negative value?

**10** Determine the $y$-intercept of all exponential functions of the form:

a $y = a^x$

b $y = a^{-x}$

c $y = -a^x$

d $y = -a^{-x}$

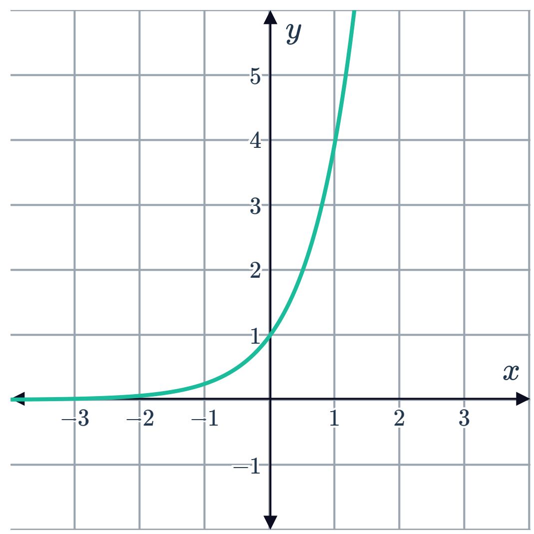
**11** Consider an exponential curve with equation of the form $y = a^x$, where $a > 0$.

- **a** What is the $y$-value of the $y$-intercept of all exponential curves of this form?
- **b** What is the $y$-value of the $y$-intercept of $y = (3.5)^x$?

**12** Consider the expression $3^x$.

- **a** Evaluate the expression when $x = -4$.
- **b** Evaluate the expression when $x = 0$.
- **c** Evaluate the expression when $x = 4$.
- **d** What happens to the value of $3^x$ as $x$ gets larger?
- **e** What happens to the value of $3^x$ as $x$ gets smaller?

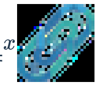
**13** Consider the graph of the function $y = 4^x$:

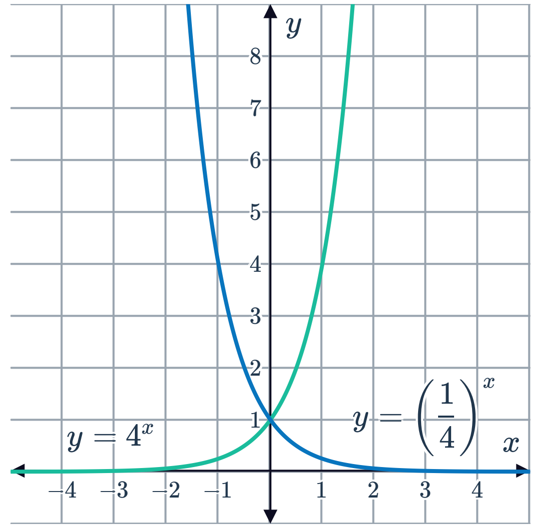



- **a** Is each $y$-value of the function positive or negative?
- **b** State the value of $y$ the graph approaches but does not reach.
- **c** State the equation and name of the horizontal line, which $y = 4^x$ gets closer and closer to but never intersects.

**14** Given the function $y = -10^x$, what is the largest possible function value? Explain your answer.

**15** Consider the graphs of the functions $y = 4^x$ and $y = \left(\frac{1}{4}\right)^x$.






**a** Which function is an increasing function?

**b** How would you describe the rate of increase of $y = 4^x$?

**16** Of the two functions $y = 4^x$ and $y = 5^x$, which increases more rapidly for $x > 0$?

**17** Consider the two functions $y = 4^{-x}$ and $y = 5^{-x}$. Which function decreases more rapidly for $x > 0$?

**18** Consider the function $y = 3^x$.

**a** Complete the table of values:

| $x$ | $-5$ | $-4$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ | $5$ | $10$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $y$ | | | | | | | | | | | | |

**b** Describe the behavior of the function as $x$ increases.

**c** State the domain of the function.

**d** State the range of the function.

**19** Consider the function $y = 2^{-x}$.

a Complete the table of values:

| $x$ | $-5$ | $-4$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ | $5$ | $10$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $y$ | | | | | | | | | | | | |

b Describe the behavior of the function as $x$ increases.

c Find the $y$-value of the $y$-intercept of the function.

d State the domain of the function.

e State the range of the function.

**20** Consider the function $y = -\left(2^x\right)$.

a Complete the table of values:

| $x$ | $-5$ | $-4$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ | $5$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $y$ | | | | | | | | | | | |

b Can the value of $y$ ever be zero or positive? Explain your answer.

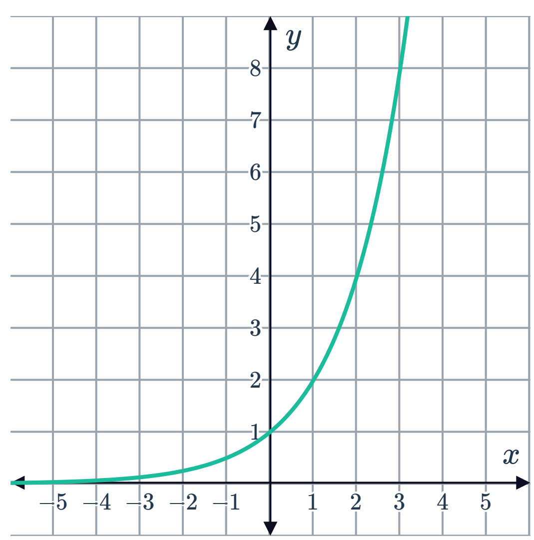
c Is $y = -\left(2^x\right)$ an increasing or decreasing function?

d Describe the behavior of the function as $x$ increases.

e State the domain of the function.

f State the range of the function.

**21** Consider the graph of the function $y = 2^x$:




a State the $y$-intercept of this graph.

b Does the graph have an $x$-intercept?

c State the domain of the function.

d State the range of the function.

e Find the value of $y$ when $x = 7$.

f Find the value of $x$ when $y = 256$.

**22** Consider the functions $y = 2^{-x}$, $y = 3^{-x}$ an[redacted]$5^{-x}$.

**a** State a characteristic that all the graphs of the functions have in common.

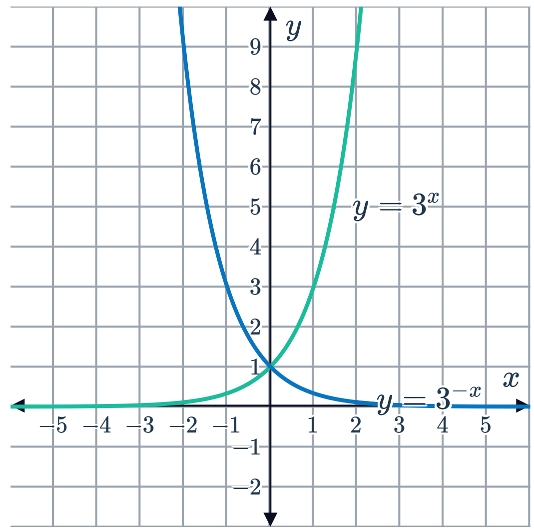
**b** What is the $y$-value of the shared $y$-intercept?

**c** Describe the nature of these functions for large values of $x$.

**23** Consider the graph of the following functions $y = 3^x$ and $y = 3^{-x}$:

**a** State the coordinates of the point of intersection of the two curves.

**b** Describe the behavior of both these functions for large values of $x$.



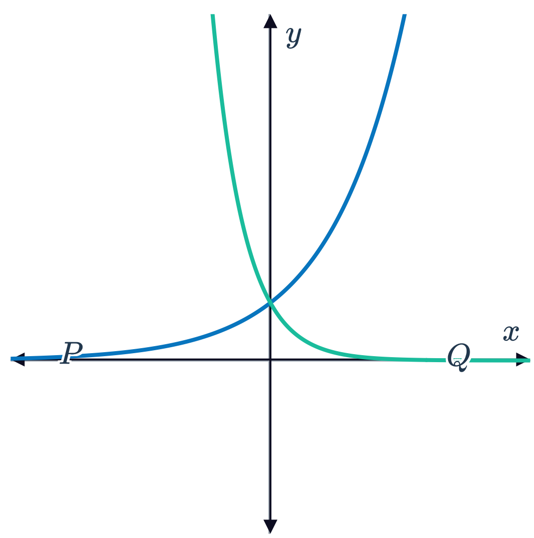

**24** Consider the given graphs of the two exponential functions $P$ and $Q$:
State whether the following pairs of equations could be the equations of the graphs $P$ and $Q$:

**a** $P : y = 2^x$
$Q : y = 2^{-x}$

**b** $P : y = (3.5)^x$
$Q : y = 6^{-x}$

**c** $P : y = 2^x$
$Q : y = 5^{-x}$

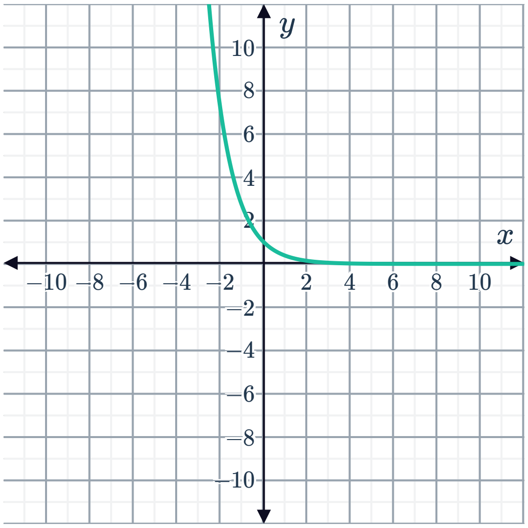
**d** $P : y = 5^x$
$Q : y = 2^{-x}$



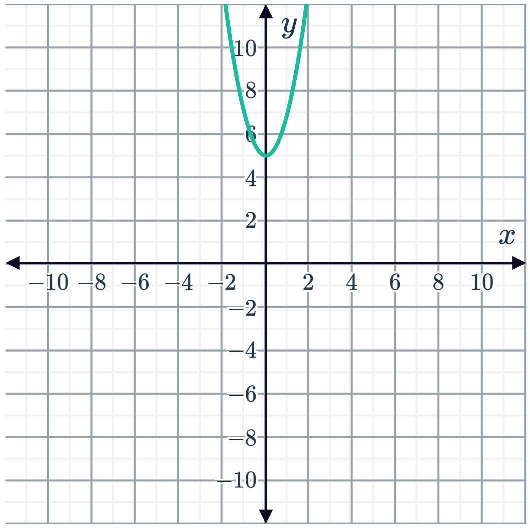
## Graphs of exponential functions

**25** Determine whether the following graphs show an exponential function:

a



b



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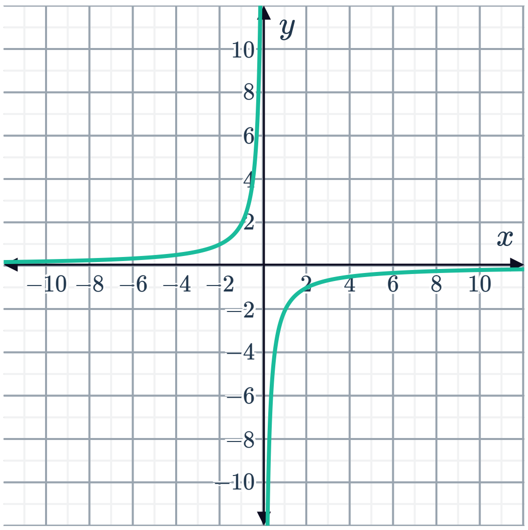
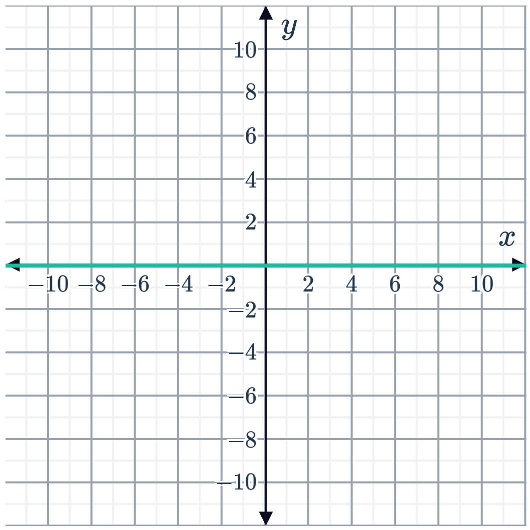
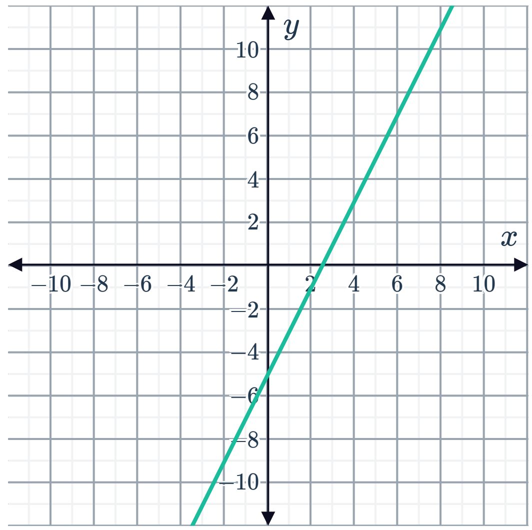
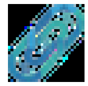


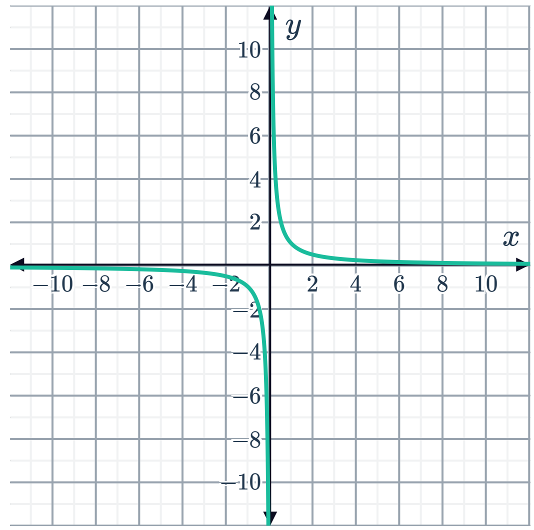
d



e







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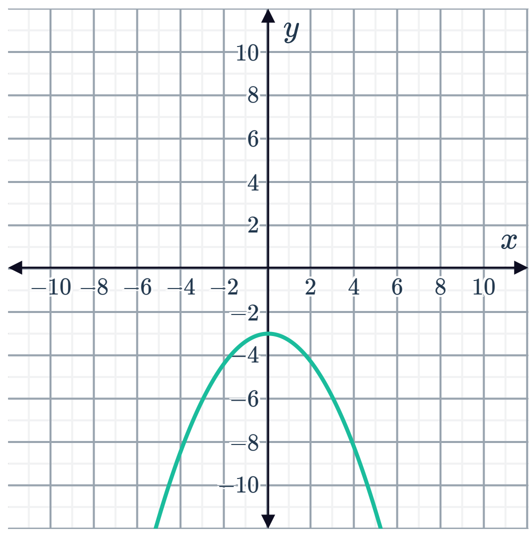

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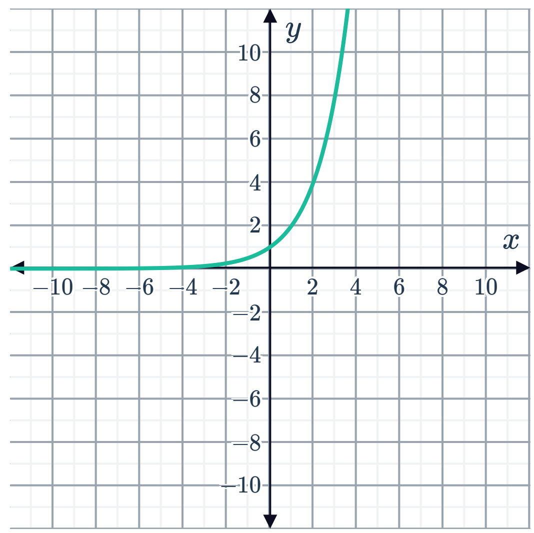


**26** Consider the equation $y = 3^x$.

**a** Complete the table of solutions for the given equation.

**b** Plot the points $C$, $D$, and $E$ on a coordinate plane.

**c** Sketch the curve that results from the entire set of solutions for the equation being graphed.

| Point | Coordinates |
|---|---|
| $A$ | $(-3, \square)$ |
| $B$ | $(-2, \square)$ |
| $C$ | $(-1, \square)$ |
| $D$ | $(\square, 1)$ |
| $E$ | $(1, \square)$ |
| $F$ | $(2, \square)$ |
| $G$ | $(3, \square)$ |

**27** Consider the equation $y = 2^x$.

a Complete the table of values:

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $y$ | | | | | | | |

b Sketch the graph of the function.

c What happens to the $y$-values as $x$ increases?

d What happens to the $y$-values as $x$ decreases?

e Does the curve crosses the $x$-axis?

f At what value of $y$ does the graph cross the $y$-axis?

**28** Consider the equation $y = 2^{-x}$.

a Complete the table of values:

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $y$ | | | | | | | |

b Sketch the graph of the function.

c What happens to the $y$-values as $x$ increases?

d What happens to the $y$-values as $x$ decreases?

e Does the curve crosses the $x$-axis?

f At what value of $y$ does the graph cross the $y$-axis?

**29** For each of the following equations:

i Find the $y$-intercept of the curve.

ii Complete the table of values:

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $y$ | | | | | | | |

iii Find the horizontal asymptote of the curve.

iv Sketch the graph of the function.

a $y = 6^x$  b $y = 3^x$  c $y = 10^x$  d $y = 3^{-x}$

30 Consider the function $y = 6^x$.

a Can the value of $y$ ever be zero or negative? Explain your answer.

b As the values of $x$ get larger and larger, what value does $y$ approach?

c As the values of $x$ get smaller and smaller, what value does $y$ approach?

d Find the $y$-value of the $y$-intercept of the curve.

e How many $x$-intercepts does the curve have?

f Sketch the graph of the function.

31 Consider the function $y = 8^{-x}$.

a Can the value of $y$ ever be zero or negative? Explain your answer.

b As the value of $x$ gets larger and larger, what value does $y$ approach?

c As the value of $x$ gets smaller and smaller, what value does $y$ approach?

d Find the $y$-value of the $y$-intercept of the curve.

e How many $x$-intercepts does the curve have?

f Sketch the graph of the function.

32 Consider the function $y = -3^x$.

a Find the $y$-intercept of the curve $y = -3^x$.

b Find the horizontal asymptote of the curve $y = -3^x$.

c Now, sketch the graph of the function.

d Is the function $y = -3^x$, an increasing or decreasing function?

33 Consider the function $y = -6^x$.

a Find the $y$-intercept of the curve $y = -6^x$.

b Find the horizontal asymptote of the curve $y = -6^x$.

c Now, sketch the graph of the function.

d Is the function $y = -6^x$, an increasing or decreasing function?

**34** Consider the function $y = -3^{-x}$.

a Find the $y$-intercept of the curve $y = -3^{-x}$.

b Find the horizontal asymptote of the curve $y = -3^{-x}$.

c Now, sketch the graph of the function.

d Is the function $y = -3^{-x}$, an increasing or decreasing function?

**35** Consider the function $y = \left(\frac{1}{2}\right)^x$.

a Determine whether the following functions are equivalent to $y = \left(\frac{1}{2}\right)^x$:

i $y = \frac{1}{2^x}$ ii $y = 2^{-x}$ iii $y = -2^x$ iv $y = -2^{-x}$

b Describe a transformation that would obtain the graph of $y = \left(\frac{1}{2}\right)^x$ from the graph of $y = 2^x$.

c Graph the functions $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$ on the same coordinate plane.

**36** Consider the function $y = \left(\frac{1}{3}\right)^x$.

a Rewrite the function in the form $y = k^{-x}$.

b Describe a transformation that would obtain the graph of $y = \left(\frac{1}{3}\right)^x$ from the graph of $y = 3^x$.

c Graph the functions $y = 3^x$ and $y = \left(\frac{1}{3}\right)^x$ on the same coordinate plane.

**37** Consider the exponentials $y = 3^x$ and $y = 2^x$.

**a** Find the equation of the horizontal asymptote of the curve $y = 3^x$.

**b** Find the equation of the horizontal asymptote of the curve $y = 2^x$.

**c** Complete the table of values for $y = 3^x$:

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $y$ | | | | | | | |

**d** Complete the table of values for $y = 2^x$:

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $y$ | | | | | | | |

**e** Now, sketch the graphs of $y = 3^x$ and $y = 2^x$ on the same coordinate plane.

**f** State the coordinates of the point at which the two graphs intersect.

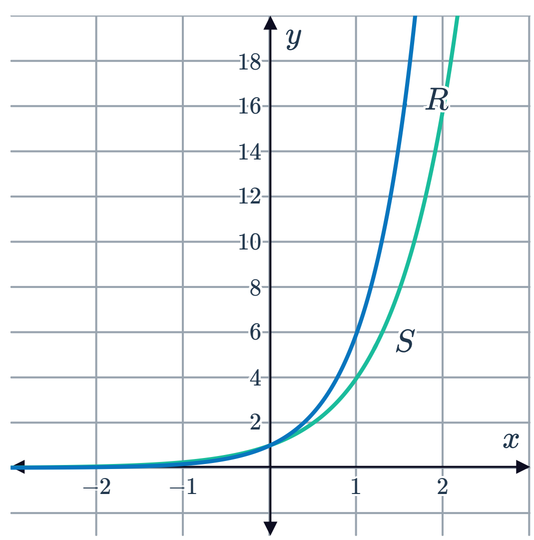
**38** Consider the functions $y = 2^x$, $y = 3^x$ and $y = 5^x$.

**a** Sketch the three functions on the same coordinate plane.

**b** State the $y$-intercept of each curve.

**c** Describe the nature of these functions for increasingly large values of $x$.

**39** Consider the graphs of the two exponential functions $R$ and $S$:




**a** One of the graphs is of $y = 4^x$ and the other graph is of $y = 6^x$.
Which is the graph of $y = 6^x$?

**b** For $x < 0$, is the graph of $y = 6^x$ above or below the graph of $y = 4^x$. Explain your answer.

**40** Consider the equation $y = -10^x$.

**a** Jenny thinks she has found a set of solutions for the equation as shown in the table:

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|
| $y$ | $-\frac{1}{100}$ | $-\frac{1}{10}$ | $-1$ | $-10$ | $-100$ | $-1000$ |

She notices that all the $y$-values are negative and concludes that for any value of $x$, $y$ must always be negative. Is she correct? Explain your answer.

**b** Sketch the graph of $y = -10^x$.

**c** Find the values of $x$ for which $y = 0$.